Recursion. Partitions and the Stirling numbers

1. Recursion

Def. The recursion method consists of reducing a problem to one or more problems of the same type, but of a smaller size / complexity, establishing a recurrence formula, computing initial values (base cases) and solving either step-by-step or finding an explicit formula.

ex. 1) Bit strings without adjacent zeros:

Case 1: [1] [n-1 bit string]
Case 2: [0] [1] [n-2 bit string]

$\Rightarrow a_n = a_{n-1} + a_{n-2}$

Obs. The recursion method:

1. We state the general problem
2. We make the first step of construction: each possible choice gives a case and what remains to be built is the same problem, but of smaller order
3. the sum rule (and, if needed, the product rule) gives the recurrence formula.
4. we find the initial values: the small cases, counted directly.
5. we compute the value one by one or look for an explicit formula.

ex. 2) The region determined by lines:

$R_n = R_{n-1} + n$

Two-dimensional recurrence:

ex. 1) Pascal's triangle:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

2) Rooks on a triangle board:

$$T(n,k) = T(n-1,k) + (n-k+1)\,T(n-1,k-1)$$

2. Partitions and the Stirling numbers:

Def. The notation $\left\{ {n \atop k} \right\}$ (the Stirling numbers of the second kind) is the number of the partitions of a set with n elements into k blocks and is read "n subset k".

Special values:

- $\left\{ {n \atop n} \right\} = 1$
- $\left\{ {n \atop 1} \right\} = 1$
- $\left\{ {n \atop k} \right\} = 0,\ k > n$
- $\left\{ {n \atop 0} \right\} = 0,\ n \geq 1$
- $\left\{ {0 \atop 0} \right\} = 1$
- $\left\{ {n \atop n-1} \right\} = \binom{n}{2}$

$$\left\{ {n \atop 2} \right\} = 2^{n-1} - 1$$

$$\left\{ {n \atop n-2} \right\} = \binom{n}{3} + 3\binom{n}{4}$$

$$\left\{ {n \atop 3} \right\} = \frac{3^{n-1} - 2^n + 1}{2}$$

The recurrence formula:

$$\left\{ {n \atop k} \right\} = \left\{ {n-1 \atop k-1} \right\} + k \cdot \left\{ {n-1 \atop k} \right\}$$

The number of surjective functions:

$$k! \cdot \left\{ {n \atop k} \right\}$$